

WOMEN'S CLOTHING E-COMMERCE ANALYSIS

OBJECTIVE:

*Analyze women’s clothing e-commerce data using Python. *Understand product sales, pricing and discounts. *Examine ratings, customer reviews and return rates. *Analyze stock levels and product characteristics. *Identify patterns and generate business insights.

1. Data Loading and Initial Overview

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:

```
df = pd.read_csv("women_clothing_50k.csv")
```

In [4]:

```
df
```

Out[4]:

	product_id	category	subcategory	material	color	size	season	style
0	WC000001	Jeans	Skinny Jeans	Denim	Red	M	Summer	Business Casual
1	WC000002	Dresses	Wrap Dress	Cotton	White	M	Winter	Casual
2	WC000003	T-Shirts	V-Neck T-Shirt	Cotton	Purple	M	All Season	Casual
3	WC000004	Sweaters	Oversized Sweater	Wool	Brown	S	Winter	Formal
4	WC000005	Sweaters	Oversized Sweater	Wool	White	M	Winter	Minimalist
...	...	...	...	...	...	...	...	...
49995	WC049996	Coats	Long Coat	Cotton	Black	L	Winter	Business Casual
49996	WC049997	Dresses	Midi Dress	Cotton Blend	White	M	Spring	Bohemian
49997	WC049998	Jeans	Wide-Leg Jeans	Cotton Blend	Red	L	Autumn	Minimalist
49998	WC049999	Sweaters	Knit Sweater	Wool	Beige	L	Autumn	Bohemian
49999	WC050000	Tops	Long Sleeve Top	Viscose	White	M	Winter	Elegant

50000 rows × 15 columns

```
In [5]: df.head()
```

```
Out[5]:
```

	product_id	category	subcategory	material	color	size	season	style	price_
0	WC000001	Jeans	Skinny Jeans	Denim	Red	M	Summer	Business Casual	3
1	WC000002	Dresses	Wrap Dress	Cotton	White	M	Winter	Casual	5
2	WC000003	T-Shirts	V-Neck T-Shirt	Cotton	Purple	M	All Season	Casual	2
3	WC000004	Sweaters	Oversized Sweater	Wool	Brown	S	Winter	Formal	7
4	WC000005	Sweaters	Oversized Sweater	Wool	White	M	Winter	Minimalist	7



```
In [7]: df.shape
```

```
Out[7]: (50000, 15)
```

```
In [8]: total_duplicates = df.duplicated().sum()
print(f"Total Duplicate Rows: {total_duplicates}")
```

Total Duplicate Rows: 0

```
In [9]: df.dtypes
```

```
Out[9]: product_id      object
category      object
subcategory    object
material       object
color          object
size           object
season         object
style          object
price_usd     float64
discount_percent  int64
rating        float64
review_count   int64
stock_quantity int64
units_sold     int64
return_rate    float64
dtype: object
```

```
In [10]: df['price_usd'].dtype
```

```
Out[10]: dtype('float64')
```

```
In [11]: df.describe()
```

Out[11]:

	price_usd	discount_percent	rating	review_count	stock_quantity	units_sold
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000
mean	61.858053	9.617800	3.056088	9.767320	37.515080	18.750000
std	36.067137	12.369801	1.905743	32.783376	39.945939	50.000000
min	12.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	34.890000	0.000000	0.000000	0.000000	9.000000	0.000000
50%	53.750000	0.000000	3.900000	2.000000	25.000000	0.000000
75%	80.030000	15.000000	4.400000	8.000000	53.000000	10.000000
max	275.560000	50.000000	5.000000	1639.000000	531.000000	312.000000



In [12]: `df.isnull().sum()`

Out[12]:

```

product_id      0
category        0
subcategory     0
material        0
color           0
size            0
season          0
style           0
price_usd       0
discount_percent 0
rating          0
review_count    0
stock_quantity  0
units_sold      0
return_rate     0
dtype: int64

```

2. Data Pre-processing

Additional Columns

In [17]: `df["discount_amount"] = df["price_usd"] * df["discount_percent"] / 100`

In [18]: `df["selling_price"] = df["price_usd"] - df["discount_amount"]`

In [20]: `df["sales_category"] = pd.cut(
 df["units_sold"],
 bins=[-1, 100, 500, np.inf],
 labels=["Low Sales", "Medium Sales", "High Sales"])`

In [21]: `df["price_category"] = pd.cut(
 df["price_usd"],
 bins=[0, 30, 70, np.inf],
 labels=["Budget", "Mid-Range", "Premium"]
)`

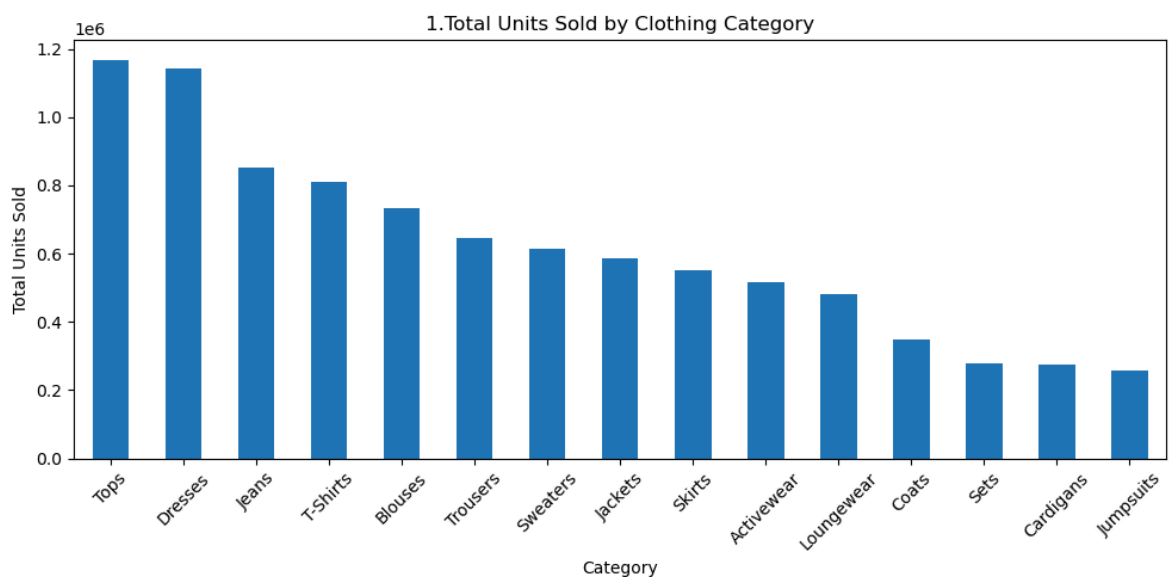
```
In [22]: df.head()
```

```
Out[22]:
```

	product_id	category	subcategory	material	color	size	season	style	price
0	WC000001	Jeans	Skinny Jeans	Denim	Red	M	Summer	Business Casual	3
1	WC000002	Dresses	Wrap Dress	Cotton	White	M	Winter	Casual	5
2	WC000003	T-Shirts	V-Neck T-Shirt	Cotton	Purple	M	All Season	Casual	2
3	WC000004	Sweaters	Oversized Sweater	Wool	Brown	S	Winter	Formal	7
4	WC000005	Sweaters	Oversized Sweater	Wool	White	M	Winter	Minimalist	7

3. Exploratory Data Analysis (EDA) & 4. Visualizations ##### # CHART 1 - Bar chart

```
In [63]: tmp = df.groupby("category")["units_sold"].sum().sort_values(ascending=False)
tmp.plot(kind="bar", figsize=(10,5))
plt.title("1.Total Units Sold by Clothing Category")
plt.xlabel("Category")
plt.ylabel("Total Units Sold")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

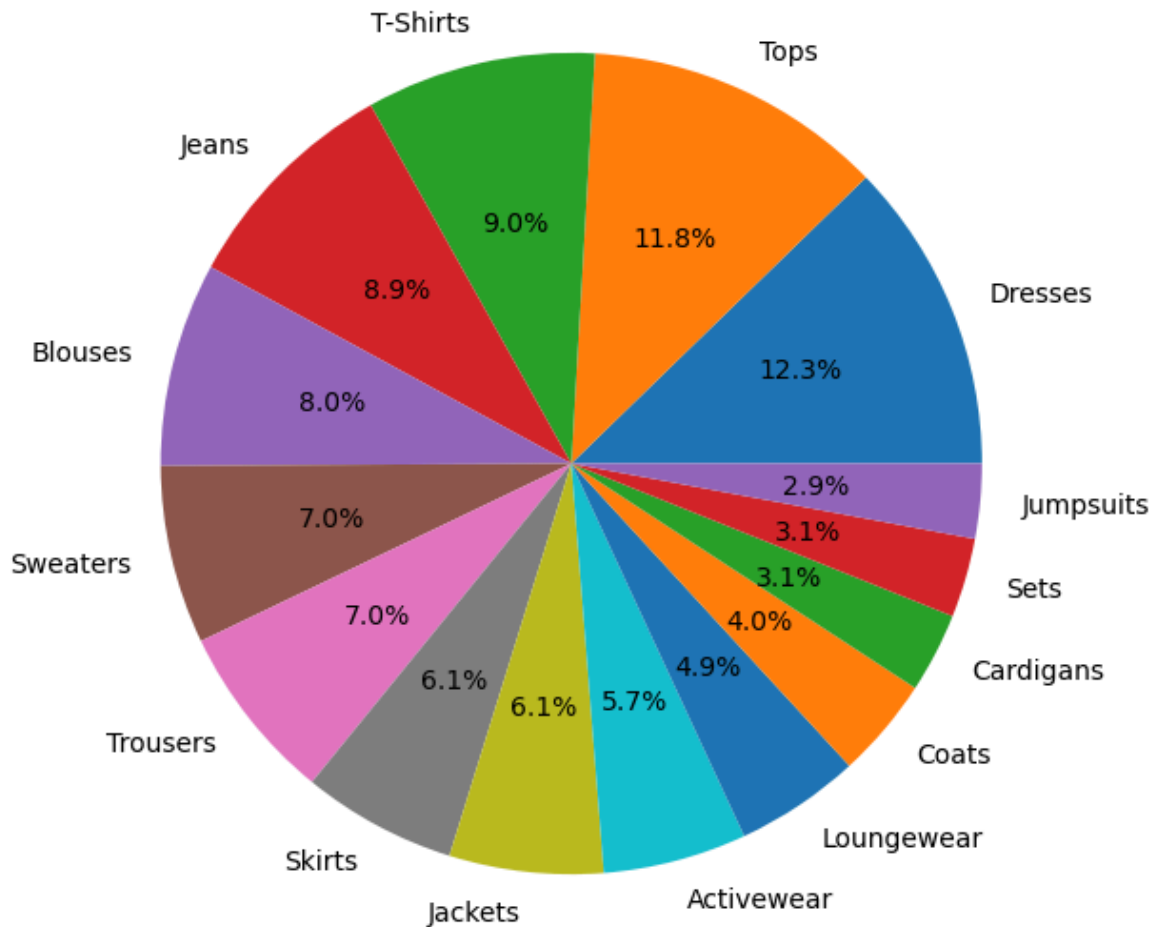


*****Tops and Dresses have the highest total units sold among the categories.*****

CHART 2 - Pie chart

```
In [61]: tmp = df["category"].value_counts()
plt.figure(figsize=(7,7))
plt.pie(tmp.values, labels=tmp.index, autopct="%1.1f%")
plt.title("2.Product Distribution by Category")
plt.show()
```

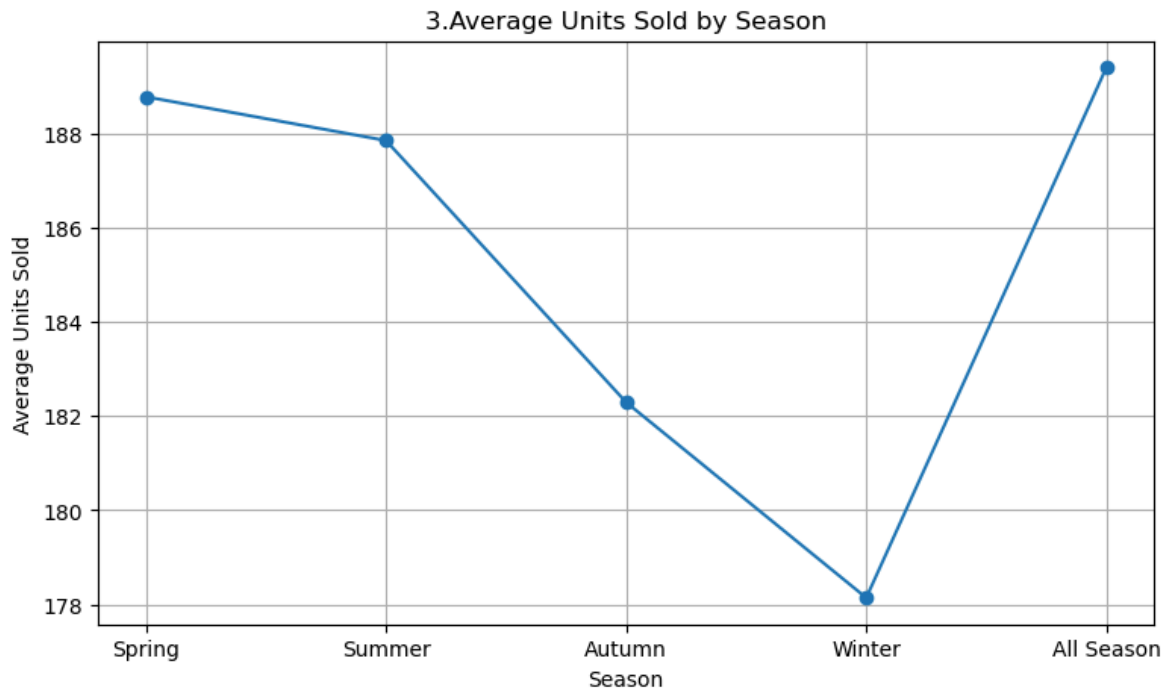
2.Product Distribution by Category



*****The product mix is spread across 15 categories; Dresses and Tops are the largest groups by record count.*****

CHART 3 - Line chart

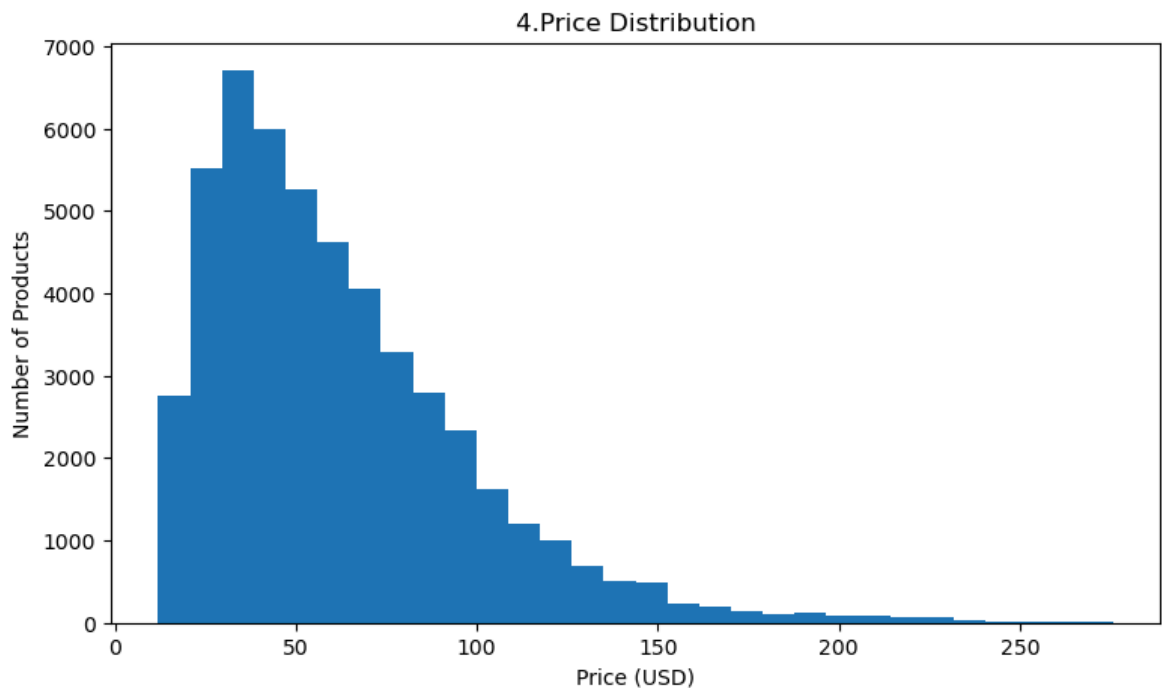
```
In [60]: season_order = ["Spring", "Summer", "Autumn", "Winter", "All Season"]
tmp = df.groupby("season", observed=True)["units_sold"].mean().reindex(season_order)
plt.figure(figsize=(9,5))
plt.plot(tmp.index, tmp.values, marker="o")
plt.title("3.Average Units Sold by Season")
plt.xlabel("Season")
plt.ylabel("Average Units Sold")
plt.grid(True)
plt.show()
```



*****Spring shows the highest average sales among the listed seasons in this dataset.*****

CHART 4 - Histogram

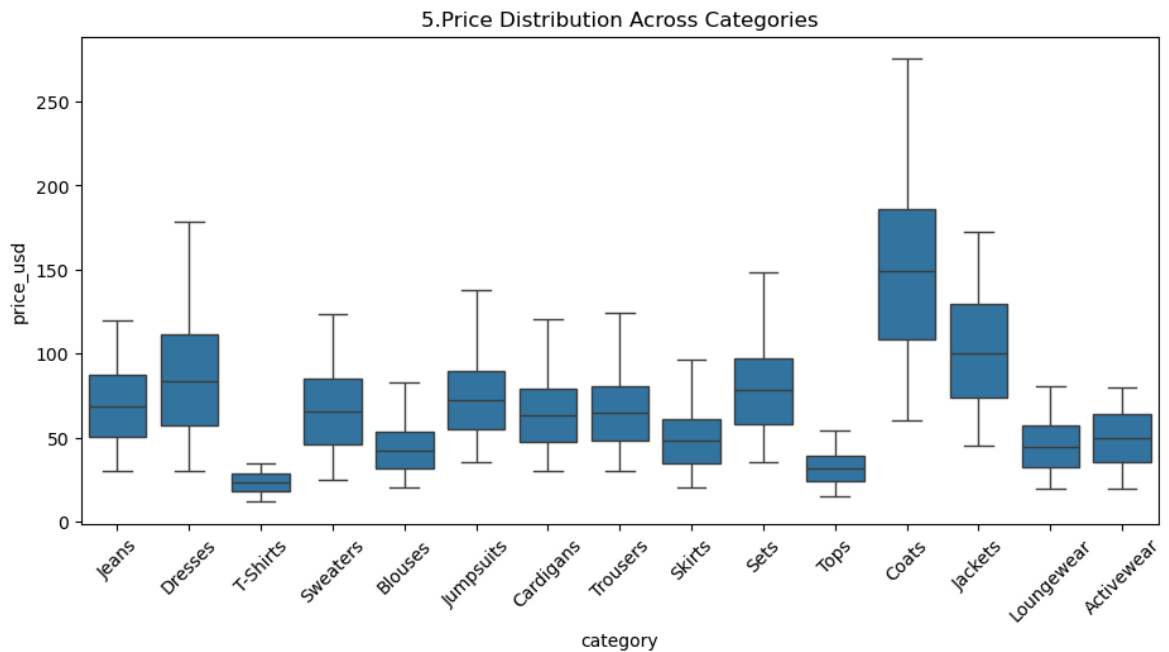
```
In [59]: plt.figure(figsize=(9,5))
plt.hist(df["price_usd"], bins=30)
plt.title("4.Price Distribution")
plt.xlabel("Price (USD)")
plt.ylabel("Number of Products")
plt.show()
```



*****Prices are concentrated toward the lower-to-middle range, with a smaller number of higher-priced products.*****

CHART 5 - Box plot

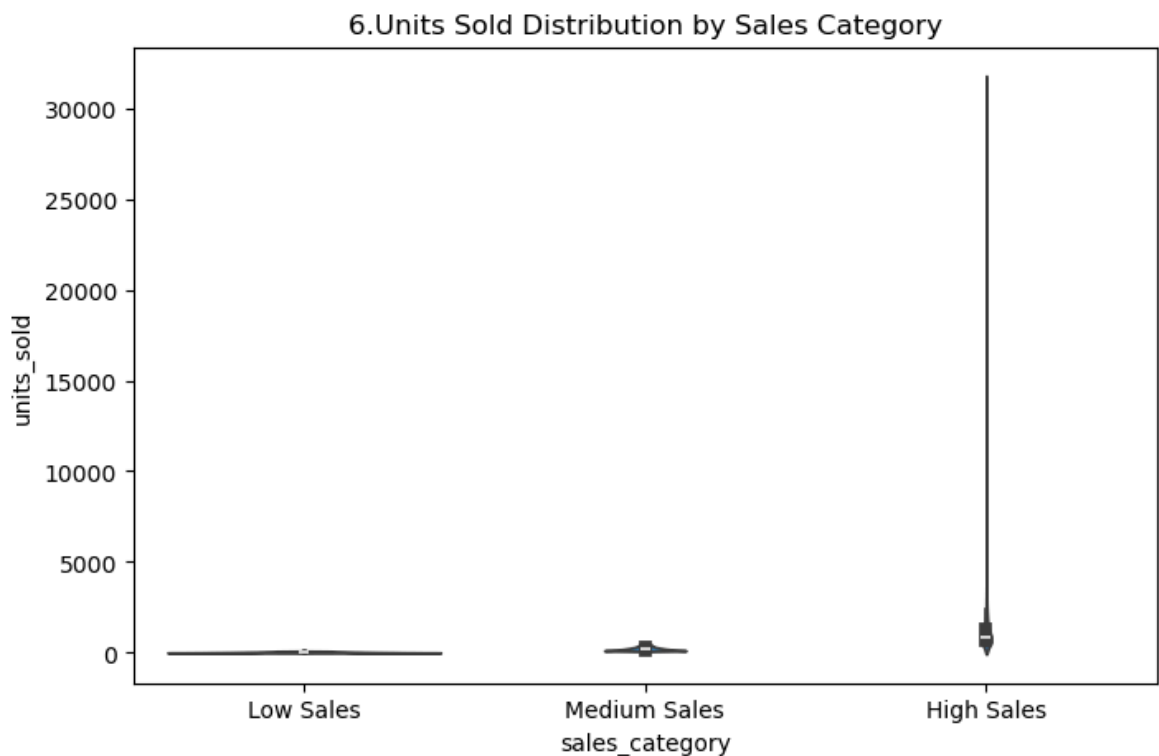
```
In [58]: plt.figure(figsize=(11,5))
sns.boxplot(data=df, x="category", y="price_usd")
plt.title("5.Price Distribution Across Categories")
plt.xticks(rotation=45)
plt.show()
```



*****Price levels vary substantially across clothing categories; higher-priced categories show wider distributions.****

CHART 6 - Violin plot

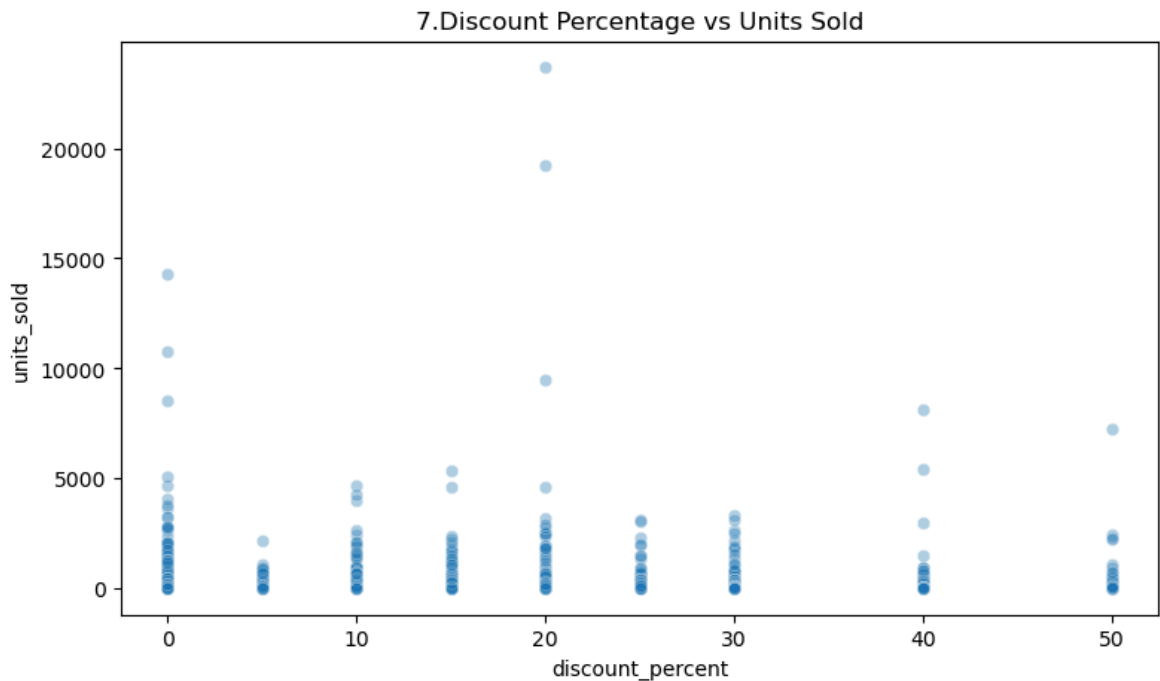
```
In [57]: plt.figure(figsize=(8,5))
sns.violinplot(data=df, x="sales_category", y="units_sold")
plt.title("6.Units Sold Distribution by Sales Category")
plt.show()
```



*****The violin plot shows the distribution of sales across the defined low, medium and high sales groups.****

CHART 7 - Scatter plot

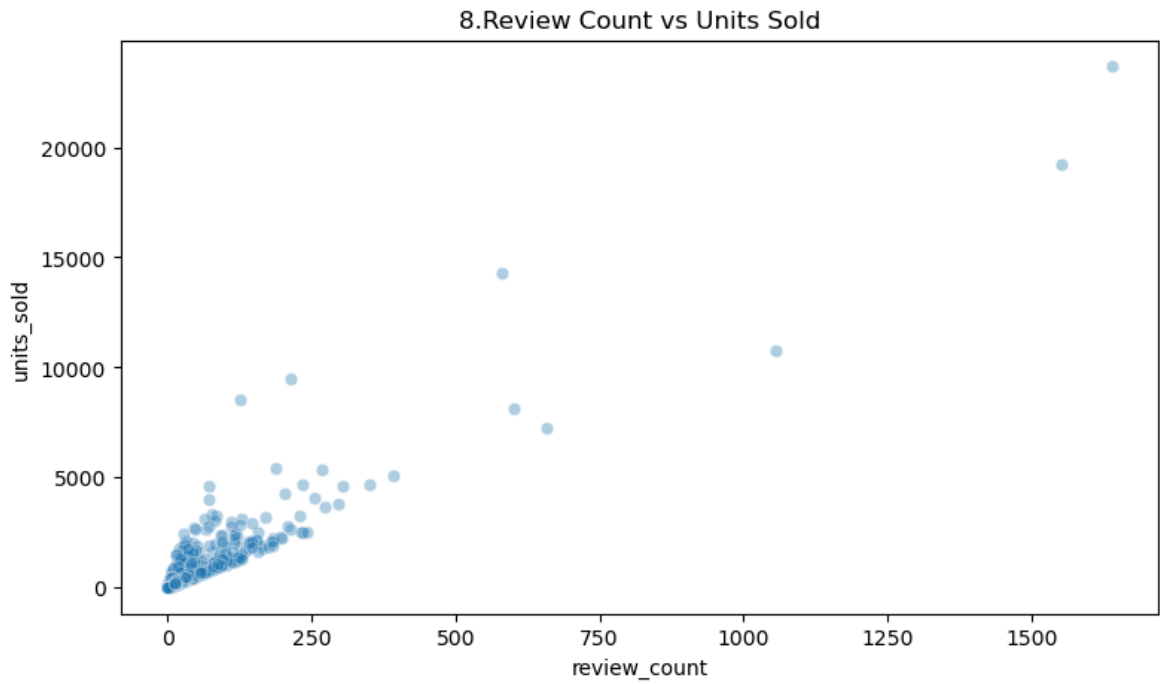
```
In [56]: sample_df = df.sample(min(5000, len(df)), random_state=42)
plt.figure(figsize=(9,5))
sns.scatterplot(data=sample_df, x="discount_percent", y="units_sold", alpha=0.35)
plt.title("7.Discount Percentage vs Units Sold")
plt.show()
```



*****Discount percentage has only a weak positive relationship with units sold; discount alone does not explain most sales variation.*****

CHART 8 - Scatter plot

```
In [55]: plt.figure(figsize=(9,5))
sns.scatterplot(data=sample_df, x="review_count", y="units_sold", alpha=0.35)
plt.title("8.Review Count vs Units Sold")
plt.show()
```

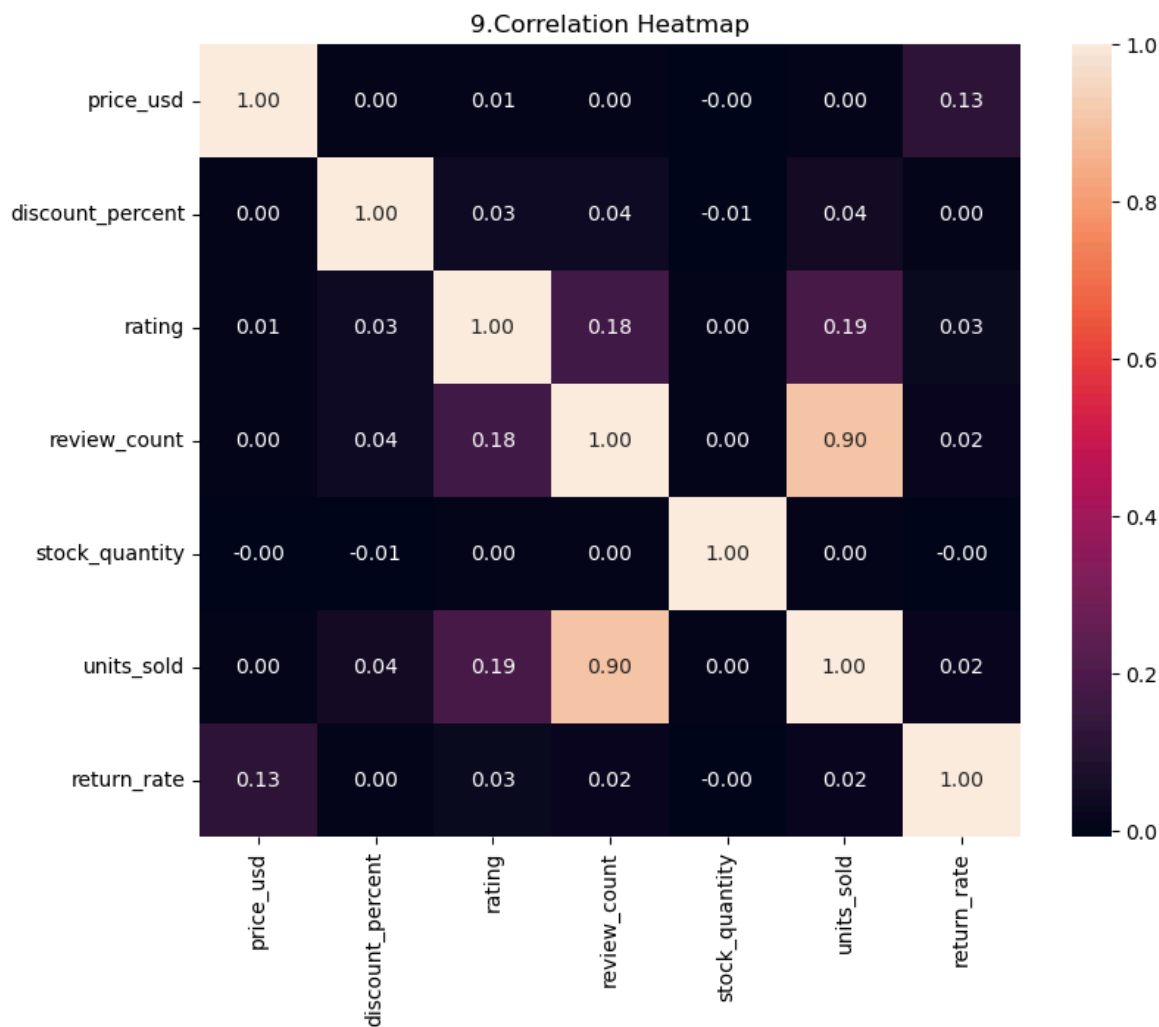


****Review count has a strong positive association with units sold; this is correlation, not proof of causation.****

CHART 9 - Heatmap

```
In [54]: corr = df[[
    "price_usd", "discount_percent", "rating",
    "review_count", "stock_quantity", "units_sold",
    "return_rate"
]].corr()

plt.figure(figsize=(9,7))
sns.heatmap(corr, annot=True, fmt=".2f")
plt.title("9. Correlation Heatmap")
plt.show()
```



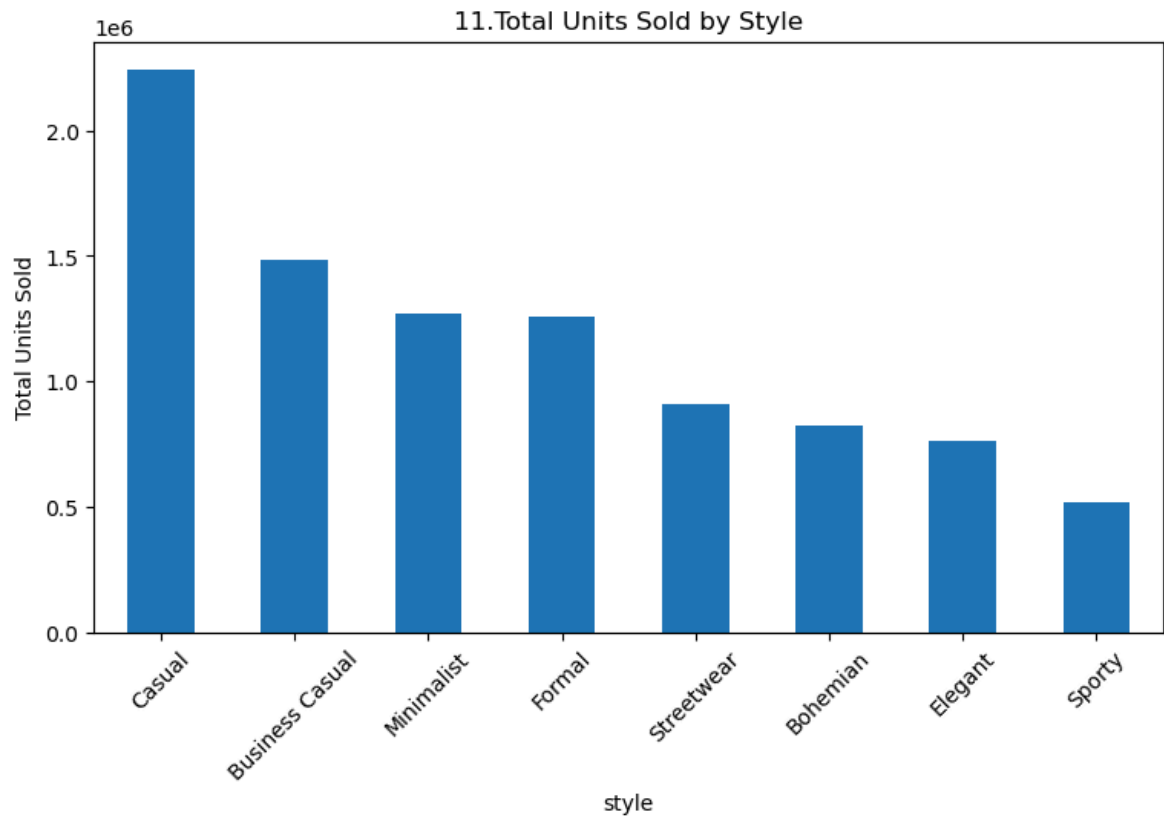
****The heatmap shows review_count has the strongest simple correlation with units_sold among the numeric variables shown.****

```
# CHART 10 - Material bar chart
tmp = df.groupby("material")["units_sold"].sum().sort_values(ascending=False)
tmp.plot(kind="bar", figsize=(9,5)) plt.title("10.Total Units Sold by Material") plt.xlabel("Material") plt.ylabel("Total Units Sold") plt.xticks(rotation=45) plt.show()
```

****Cotton is the highest-volume material by total units sold.****

CHART 11 - Style bar chart

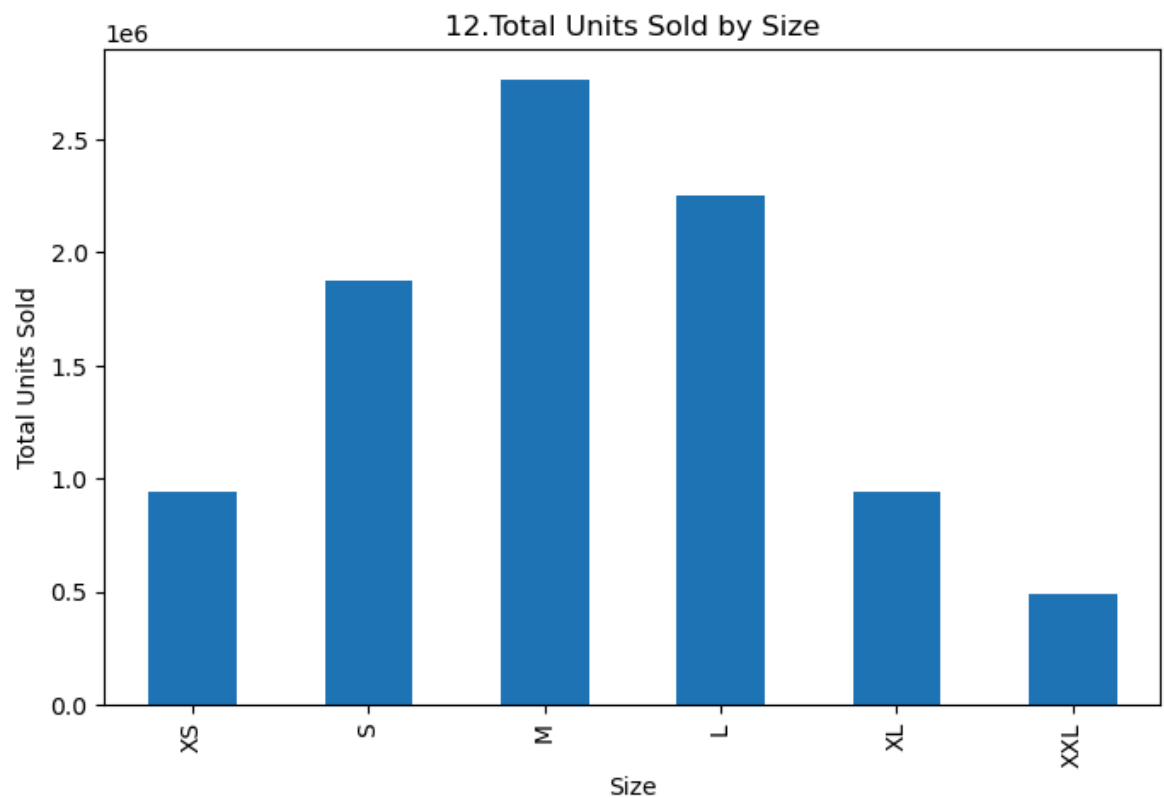
```
In [52]: tmp = df.groupby("style")["units_sold"].sum().sort_values(ascending=False)
tmp.plot(kind="bar", figsize=(9,5))
plt.title("11.Total Units Sold by Style")
plt.ylabel("Total Units Sold")
plt.xticks(rotation=45)
plt.show()
```



****Casual is the highest-volume style by total units sold.****

CHART 12 - Size bar chart

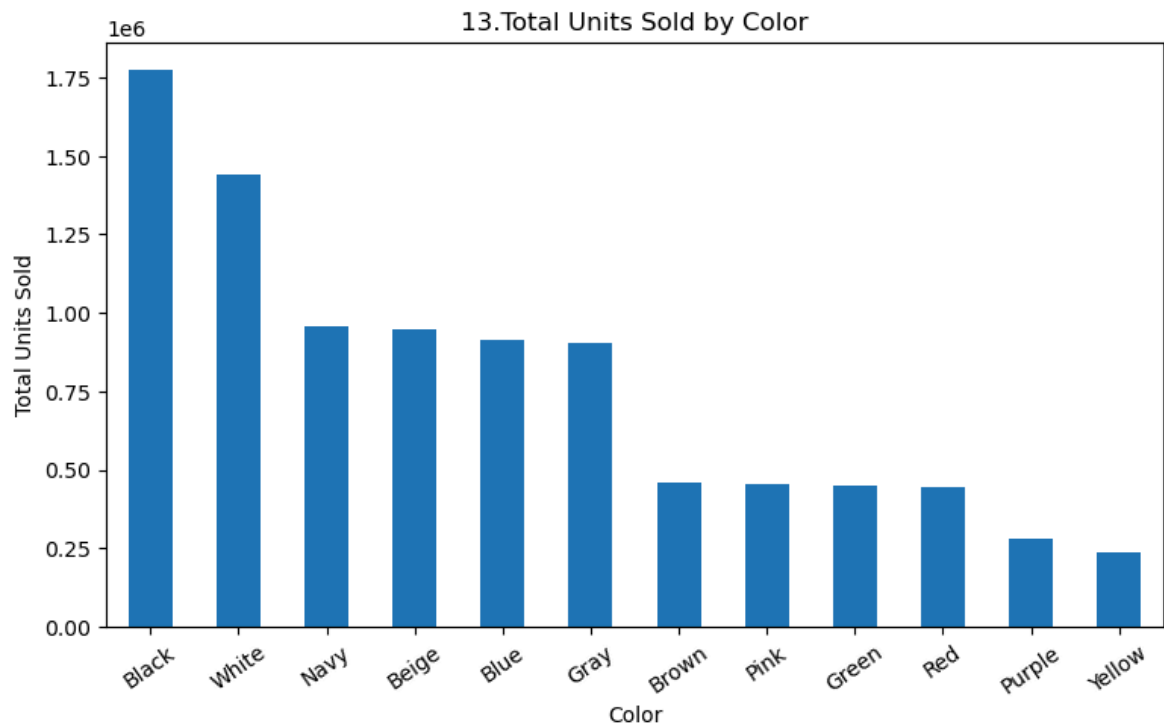
```
In [51]: tmp = df.groupby("size")["units_sold"].sum().reindex(["XS", "S", "M", "L", "XL", "XXL"])
tmp.plot(kind="bar", figsize=(8,5))
plt.title("12.Total Units Sold by Size")
plt.xlabel("Size")
plt.ylabel("Total Units Sold")
plt.show()
```



****M is the highest-volume size, followed by L.****

CHART 13 - Color bar chart

```
In [50]: tmp = df.groupby("color")["units_sold"].sum().sort_values(ascending=False)
tmp.plot(kind="bar", figsize=(9,5))
plt.title("13.Total Units Sold by Color")
plt.xlabel("Color")
plt.ylabel("Total Units Sold")
plt.xticks(rotation=35)
plt.show()
```



****Black is the highest-volume color by total units sold.****

CHART 14 - Category x Season pivot table

```
In [49]: pivot = pd.pivot_table(
    df,
    values="units_sold",
    index="category",
    columns="season",
    aggfunc="sum",
    fill_value=0
)
pivot = pivot[season_order]

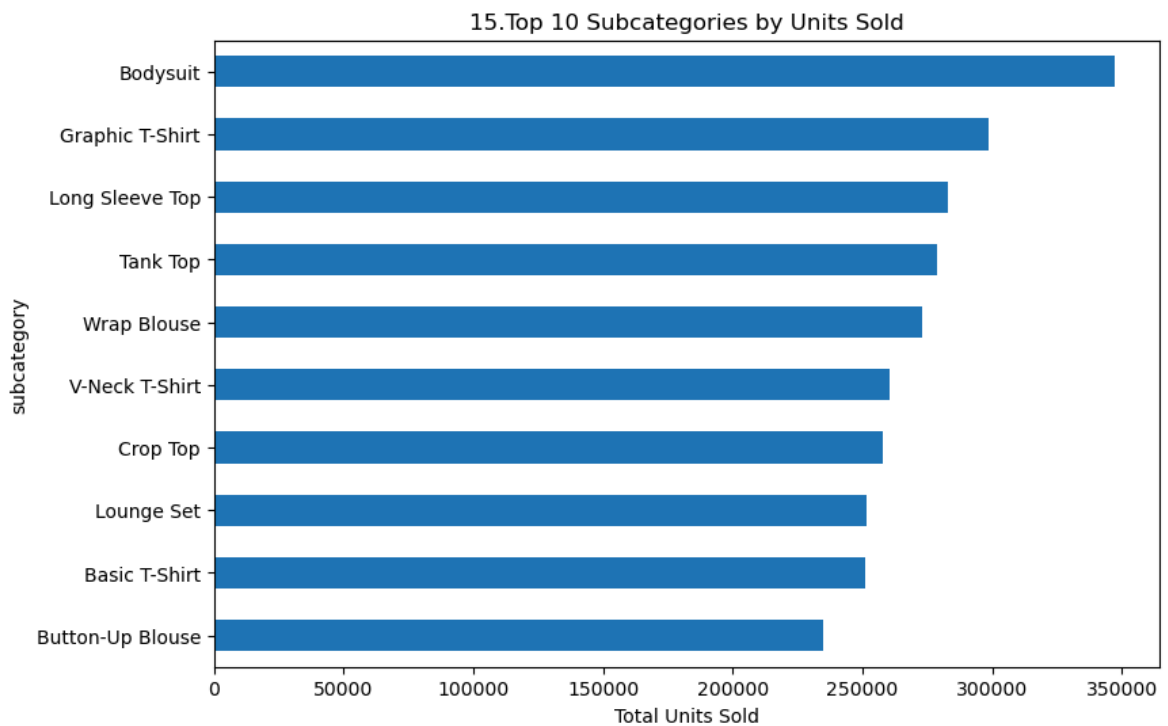
plt.figure(figsize=(10,8))
sns.heatmap(pivot, annot=True, fmt=".0f")
plt.title("14.Category x Season Sales")
plt.show()
```



****The category × season heatmap helps identify which category-season combinations contribute the most sales.****

CHART 15 - Top 10 subcategories

```
In [48]: tmp = df.groupby("subcategory")["units_sold"].sum().sort_values(ascending=False)
tmp.sort_values().plot(kind="barh", figsize=(9,6))
plt.title("15.Top 10 Subcategories by Units Sold")
plt.xlabel("Total Units Sold")
plt.show()
```



****Bodysuit is the highest-selling subcategory in this dataset.****

Correlation values####

```
In [65]: print("\nCorrelation with Units Sold:")
print(df[["price_usd", "discount_percent", "rating",
          "review_count", "stock_quantity", "return_rate",
          "units_sold"]].corr()["units_sold"].sort_values(ascending=False))
```

```
Correlation with Units Sold:
units_sold      1.000000
review_count    0.901413
rating          0.188214
discount_percent 0.044819
return_rate     0.022652
price_usd       0.001751
stock_quantity  0.001024
Name: units_sold, dtype: float64
```

#####Key groupby analysis#####

```
In [66]: print("\nSales by Category:")
print(df.groupby("category")["units_sold"].sum().sort_values(ascending=False))

print("\nSales by Season:")
print(df.groupby("season")["units_sold"].sum().sort_values(ascending=False))

print("\nSales by Material:")
print(df.groupby("material")["units_sold"].sum().sort_values(ascending=False))
```

Sales by Category:

category

Tops	1167189
Dresses	1142011
Jeans	853870
T-Shirts	810126
Blouses	733189
Trousers	645686
Sweaters	613191
Jackets	587283
Skirts	552580
Activewear	515406
Loungewear	480696
Coats	347864
Sets	277445
Cardigans	275549
Jumpsuits	257448

Name: units_sold, dtype: int64

Sales by Season:

season

Summer	1991461
Autumn	1910984
Winter	1800796
Spring	1790825
All Season	1765467

Name: units_sold, dtype: int64

Sales by Material:

material

Cotton	2766898
Cotton Blend	1518130
Polyester	1238405
Denim	1039784
Polyester Blend	697007
Viscose	625887
Wool	574596
Linen	411971
Nylon	211059
Leather	175796

Name: units_sold, dtype: int64

**** 5.Insight Generation and Report ****

INSIGHTS

1.Tops recorded the highest total units sold , followed closely by Dresses . 2.Cotton was the highest-volume material with units sold. 3.Casual was the highest-volume style with units sold. 4.Size M had the highest total units sold followed by L . 5.Black was the highest-volume color with units sold. 6.Summer recorded the highest total units sold among the seasons . 7.The strongest simple numeric relationship is review_count vs units_sold (correlation ≈ 0.90); this indicates association, not proof that reviews cause sales.

CONCLUSION

The analysis shows clear differences in sales across categories, materials, styles, sizes, colors and seasons. Tops, Dresses, Cotton, Casual style and size M are among the strongest volume segments in this dataset. Review count has a strong association with units sold, while discount percentage shows only a weak association.

